INCH-POUND

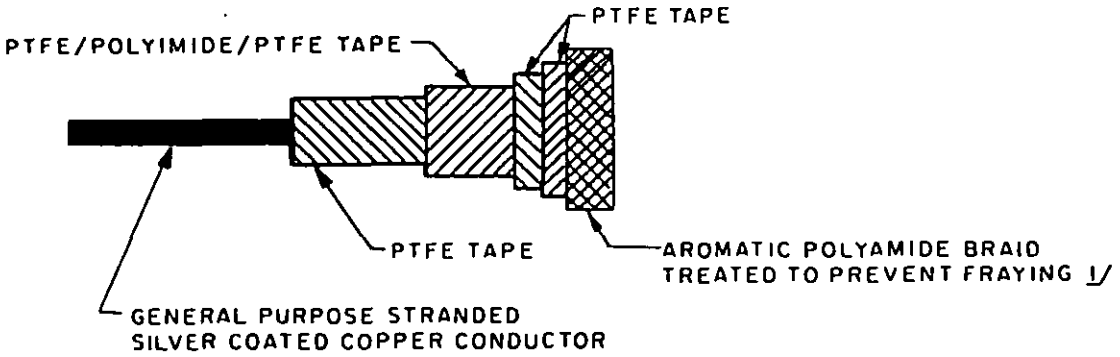
MIL-W-22759/83
9 September 1994

MILITARY SPECIFICATION SHEET

WIRE, ELECTRICAL, POLYTETRAFLUOROETHYLENE/POLYIMIDE INSULATED, NORMAL WEIGHT, SILVER COATED COPPER CONDUCTOR, 200°C, 600-VOLT

This specification sheet has been approved for use by the all departments and agencies within the Department of Defense; however, it is not intended for use in Naval aircraft or Naval Air Systems applications.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-W-22759.



PTFE-POLYTETRAFLUOROETHYLENE

1/ Braid: Bright aromatic polyamide yarn, 200 Denier, 100 filaments, tightly formed, uniform in appearance, treated with a clear finisher coating. The finisher coating shall be compatible with the temperature rating and performance requirements of the insulated wire.

FIGURE 1. General configuration.

AMSC N/A FSC 6145

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TABLE I. Construction details.

Part No. 1/	Wire Size	Conductor			Finished Wire			
		Stranding (Number of Strands X AWG Gauge of Strands)	Diameter (in.)		Resistance at 20°C (68°F) (ohms/ 1000 ft) (Max)	Diameter (in.)		Weight (lb/ 1000 ft) (Max)
			Min	Max		Min	Max	
M22759/ -2- *	2	665 x 30	0.320	0.340	0.170	0.360	0.380	227.0
M22759/ -1- *	1	817 x 30	0.360	0.380	0.139	0.400	0.420	295.0
M22759/ -01-*	0	1045 x 30	0.405	0.425	0.108	0.442	0.462	346.0
M22759/ -02-*	00	1330 x 30	0.440	0.475	0.085	0.498	0.528	432.0
M22759/ -03-*	000	1665 x 30	0.500	0.540	0.068	0.554	0.584	541.0
M22759/ -04-*	0000	2109 x 30	0.565	0.605	0.054	0.615	0.655	679.0

1/ Part number: The preferred color is dark green with the color designator 5D. Example: Size 2 dark green - M22759/ -2-5D.

TABLE II. Wire insulation materials. 1/

Tape Code	Thickness (Nominal)	Material
1	0.002	0.0005(PTFE)/0.001(Polyimide)/0.0005(PTFE)
2	0.002	PTFE (Skived)
3	0.003	PTFE (Unsintered or presintered bondable)

1/ Physical properties of PTFE tapes (skived and unsintered) shall be in accordance with MIL-W-22759 requirements.

TABLE III. Physical properties of PTFE/Polyimide/PTFE tape.

Tensile strength	12,000 lb/in^2 (average minimum)
Elongation	40 percent (average minimum)
Dielectric strength	3,000 volts/mil (average minimum)

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TABLE IV. Tape overlap requirements. 1/

Wire Size	Wrap 1		Wrap 2		Wrap 3		Wrap 4		Nominal Wall Thickness
	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	Tape Code	Percent Overlap (minimum)	
2	2	20	1	50	3	50	3	50	.017
1	2	20	1	50	3	50	3	50	.017
0	2	20	1	50	3	50	3	50	.017
02	2	20	1	50	3	50	3	50	.017
03	2	20	1	50	3	50	3	50	.017
04	2	20	1	50	3	50	3	50	.017

1/ Wrap 1 is innermost tape which is in contact with the conductor. Wraps 2, 3, and 4 are progressively further away from the conductor core.

RATINGS:

Temperature rating: 200°C (392°F) maximum continuous conductor temperature.
Voltage rating: 600 Volts (rms) at sea level.

ADDITIONAL REQUIREMENTS:

Wet arc propagation resistance (Test required for initial qualification only): Qualification by similarity to MIL-W-22759/TT-20.

Dry arc propagation resistance (Test required for initial qualification only): Qualification by similarity to MIL-W-22759/TT-20.

Blocking: 200°C ± 2°C (392°F ± 3.6°F).

Color: For braided constructions, preferred color shall be dark green with Munsell color limits of 5Y 3/2 and 5B 2/0.5. Conformity of color to the limits of MIL-STD-104 shall not be required after oven exposure.

Color striping or banding durability: Not required.

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Conductor strand adhesion: Required.

Continuous lengths: Schedule B.

Dynamic cut-through (Test required for initial qualification only): Qualification by similarity to M22759/TT-20.

Flammability: Test in accordance with MIL-STD-2223 method 1006 procedure A. Requirements:

Duration of after-flame	3 seconds (max)
Flame travel	3.0 inches (max)
No flaming of tissue	

High-frequency spark test: (When used in lieu of Impulse Dielectric Test) Test in accordance with MIL-STD-2223 method 3008. 5.7 kilovolts (rms). Test 100 percent of the wire.

Humidity resistance: After humidity exposure, wire shall meet the requirements for initial insulation resistance.

Identification of product: Color code designator not required.

Identification durability: Not required.

Immersion (Test required for initial qualification only): Test in accordance with MIL-STD-2223 method 1001 including additional fluids listed in Table V of this specification. Use mandrels and weights listed in Table VI for bend testing. Dielectric test, 2500 volts (rms), 60 Hz. For the turbine fuel immersion test of MIL-STD-2223, either Type JP-4 or MIL-T-83133 Type JP-8 (NATO Type F-34) may be used.

Impulse dielectric test: 8.0 kilovolts (peak). Test 100 percent of the wire.

Insulation resistance: 3000 megohms for 1000 feet (min).

Life cycle: 500 hours at 230°C ± 2°C (446°F ± 3.6°F). Dielectric test, 2500 volts (Rms), 60 Hz. Use mandrels coated with polytetrafluoroethylene such that the diameter of the mandrels, after coating, still conform to the required test mandrel diameters in Table VI. After oven exposure, layers shall not separate and/or tapes shall not lift along the insulation or at the ends.

Low temperature (cold bend): Use mandrels and weights specified in Table VI. Chamber temperature, -65°C ± 2°C (-85°F ± 3.6°F). Dielectric test, 2500 volts (rms), 60 Hz.

Shrinkage: Maximum change in measurement, 0.125 inch. Test at 230°C ± 2°C (446°F ± 3.6°F).

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Smoke: 200°C ± 5°C (392°F ± 9°F); no visible smoke.

Solderability: Required.

Strippability: Not required.

Thermal Index (Test required for initial qualification only): Qualification by similarity to M22759/TT.

Thermal shock resistance: Oven temperature, 200°C ± 2°C (392°F ± 3.6°F). Maximum change in measurement, 0.125 inch. No cracking.

Wrap (mandrel wrap): No cracking, no dielectric breakdown. Use mandrel sizes specified in Table VI. No wrinkles shall be visible in the insulation at 3X magnification. Dielectric test 2500 volts (rms), 60 Hz.

TABLE V. Fluid table.

Test fluid	Test temperature (°C(°F))	Immersion time (hrs)
A. MIL-A-8243, Anti-icing and Deicing-Defrosting Fluid, undiluted	48-50 (118-122)	20
B. MIL-A-8243, Anti-icing and Deicing-Defrosting Fluid, diluted 60/40 (fluid/water) ratio	48-50 (118-122)	20
C. MIL-C-43616, Cleaning Compound, Aircraft Surface	48-50 (118-122)	20
D. ASTM D1153, Methyl Isobutyl Keyton (For use in Organic Coatings)	20-25 (68-77)	168
E. SAE-AS-1241, Fire Resistant Hydraulic Fluid for Aircraft	48-50 (118-122)	20
F. MIL-L-7808, Lubricating Oil, Aircraft Turbine Engine, Synthetic Base	118-121 (244-250)	30
G. MIL-C-87936, Cleaning Compound, Aircraft Surface, Alkaline Waterbase, undiluted	63-68 (145-154)	20

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TABLE V. Fluid table - Continued.

Test fluid	Test temperature (°C(°F))	Immersion time (hrs)
H. MIL-C-87936, Cleaning Compound, Aircraft Surface, Alkaline Waterbase, diluted 25/75 (fluid/ water) ratio	63-68 (145-154)	20
I. TT-S-735, Standard Test Fluids: Hydrocarbon, Type I	20-25 (68-77)	168
J. TT-S-735, Standard Test Fluids: Hydrocarbon, Type II	20-25 (68-77)	168
K. TT-S-735, Standard Test Fluids: Hydrocarbon, Type III	20-25 (68-77)	168
L. TT-S-735, Standard Test Fluids: Hydrocarbon, Type VII	20-25 (68-77)	168
M. Dielectric-coolant fluid, synthetic silicate ester base, Monsanto coolanol 25 or approved equivalent	20-25 (68-77)	168
N. MIL-G-3056, Gasoline, Automotive, Combat	20-25 (68-77)	168

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TABLE VI. Test mandrel and test load requirements.

Wire Size (AWG)	Test Mandrel Diameter 1 (in.)			Test Load 1/ (lb)	
	Cold Bend	Life Cycle/ Bend test	Wrap	Cold Bend	Life Cycle/ Bend test
2	8.00	6.00	2.00	15.00	6.00
1	10.00	8.00	2.50	15.00	6.00
0	10.00	8.00	3.00	15.00	6.00
00	12.00	10.00	4.00	20.00	8.00
000	18.00	10.00	5.00	30.00	10.00
0000	18.00	10.00	6.00	30.00	10.00

1/ Tolerance shall be ± 3 percent of the given values.

Qualification of wire:

For qualification, a source is required to submit data on quality conformance tests and any finished wire tests as required by the qualification authorization letter. All other testing will be performed by the qualifying activity at the source's expense.

Due to the extended time period over which the thermal index test is performed, a source may become qualified under this specification sheet while this test is still in progress.

Custodian:
Army - CR
Navy - AS
Air Force - 85
NASA - NA

Preparing Activity
Navy - AS

(Project 6145-2044-01)

Review Activities:
Army - AV, MI
Air Force, - 11, 19, 70, 99
DLA - ES, IS

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